Data Sciences – CentraleSupélec
Advanced Machine Learning
Course VII – Inference on Graphical Models

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From a distribution to a graph

Goal: Recover relationships between variables from observations.

$$X = (X^{(1)}, \ldots, X^{(p)})^\top \sim \mathcal{P}.$$

An undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ has vertices $\mathcal{V} = \{1, \ldots, p\}$, one for each random variable.

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- The joint distribution describes how the variables vary together.
- The graph encodes their conditional-independence structure.
- An edge is an unordered pair $\{j, k\}$, with $j \neq k$.

Our focus: Gaussian distributions and sparse undirected graphs.

Conditional independence and missing edges

Write $R = \mathcal{V} \setminus \{j, k\}$. Conditional independence means

$$p(x^{(j)}, x^{(k)} \mid x^{(R)}) = p(x^{(j)} \mid x^{(R)}) p(x^{(k)} \mid x^{(R)}).$$

Pairwise Markov property: For distinct j, k,

$$\{j, k\} \notin \mathcal{E} \quad \Longrightarrow \quad X^{(j)} \perp\!\!\!\perp X^{(k)} \mid X^{(R)}.$$

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For the **conditional-independence graph**, we define edges exactly by conditional dependence. Then the implication is an equivalence.

A missing edge means no remaining dependence once the other variables are known. It does not mean marginal independence.

Gaussian covariance and precision

Model: Throughout the population derivations, variables are centered:

$$X \sim \mathcal{N}(0, \Sigma), \qquad \Sigma = \mathbb{E}[XX^\top] \in \mathbb{R}^{p \times p}, \quad \Sigma \succ 0.$$

Positive definiteness means $v^\top \Sigma v > 0$ for every $v \neq 0$.

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Precision matrix:

$$\boxed{P = \Sigma^{-1} \succ 0}, \qquad p_X(x) = \frac{(\det P)^{1/2}}{(2\pi)^{p/2}} \exp\left(-\frac{1}{2} x^\top P x\right).$$

Covariance describes marginal associations. Precision will describe conditional associations.

The key Gaussian equivalence

For a nondegenerate Gaussian vector and distinct j, k,

$$\boxed{P_{jk} = 0 \quad \Longleftrightarrow \quad X^{(j)} \perp\!\!\!\perp X^{(k)} \mid X^{(\mathcal{V}\setminus\{j,k\})}.}$$

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Thus its conditional-independence graph has edge set

$$\mathcal{E} = \big\{\{j,k\} : 1 \le j < k \le p,\ P_{jk} \neq 0\big\}.$$

- Only off-diagonal entries encode edges.
- Symmetry $P_{jk} = P_{kj}$ gives an undirected graph.
- The diagonal entries are positive and encode conditional variances.

Learning the graph amounts to finding the off-diagonal zeros of P.

Why does the precision matrix encode the graph?

Fix $X^{(R)} = x^{(R)}$, where $R = \mathcal{V} \setminus \{j, k\}$. The conditional log-density in $x^{(j)}$ and $x^{(k)}$ contains

$$-\frac{1}{2}\left(P_{jj}(x^{(j)})^2 + 2P_{jk}x^{(j)}x^{(k)} + P_{kk}(x^{(k)})^2\right)$$

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plus linear terms and a constant.

- If $P_{jk} = 0$, it separates into a function of $x^{(j)}$ and a function of $x^{(k)}$: the conditional density factorizes.
- If $P_{jk} \neq 0$, the cross term couples the two variables.

The equivalence follows directly from the quadratic Gaussian exponent.

Partial correlation

Ordinary correlation measures marginal association:

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Partial correlation removes the linear effect of the other variables from both $X^{(j)}$ and $X^{(k)}$, then correlates the residuals. For the Gaussian model,

$$\boxed{\rho_{jk|R} = -\frac{P_{jk}}{\sqrt{P_{jj}P_{kk}}}}, \qquad R = \mathcal{V} \setminus \{j, k\}.$$

It also equals the correlation in the conditional Gaussian distribution.

In a GGM, zero partial correlation is equivalent to a missing edge.

Each node is a regression problem

Regress $X^{(j)}$ onto $X^{(-j)}$, the vector of all remaining variables:

$$X^{(j)} = \sum_{r \neq j} \beta_r^{(j)} X^{(r)} + \epsilon^{(j)}.$$

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Completing the square in the Gaussian density gives

$$X^{(j)} \mid X^{(-j)} \sim \mathcal{N}\left(-\frac{1}{P_{jj}} \sum_{r \neq j} P_{jr} X^{(r)}, \frac{1}{P_{jj}} \right).$$

Hence

$$\boxed{\beta_r^{(j)} = -\frac{P_{jr}}{P_{jj}}}, \qquad \epsilon^{(j)} \sim \mathcal{N}(0, P_{jj}^{-1}), \quad \epsilon^{(j)} \perp\!\!\!\perp X^{(-j)}.$$

Course II: regression coefficients identify relevant predictors.

From regression coefficients to neighbors

For distinct j, k,

$$\beta_k^{(j)} = -P_{jk}/P_{jj}, \qquad \beta_j^{(k)} = -P_{jk}/P_{kk}.$$

Since $P_{jj}, P_{kk} > 0$,

$$\boxed{\{j,k\} \in \mathcal{E} \iff P_{jk} \neq 0 \iff \beta_k^{(j)} \neq 0 \iff \beta_j^{(k)} \neq 0.}$$

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Neighborhood of node j:

$$S^{(j)} = \{r \neq j : \beta_r^{(j)} \neq 0\}.$$

Neighborhood recovery is variable selection in a linear regression.

Nodewise regression: one LASSO per node

Let $D \in \mathbb{R}^{n \times p}$ contain centered observations in its rows. For node j, use response $D_{:j} \in \mathbb{R}^n$ and design $D_{:-j} \in \mathbb{R}^{n \times (p-1)}$.

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$$\boxed{\widehat{\beta}^{(j)} \in \underset{b \in \mathbb{R}^{p-1}}{\operatorname{argmin}} \left\{ \frac{1}{2n} \| D_{:j} - D_{:-j} b \|_2^2 + \lambda_j \| b \|_1 \right\}.}$$

Coordinates of b retain their original node labels $r \neq j$.

$$\widehat{S}^{(j)} = \{ r \neq j : \widehat{\beta}_r^{(j)} \neq 0 \}.$$

Repeat for $j = 1, \ldots, p$. Course II's LASSO selects a sparse neighborhood.

Meinshausen and Bühlmann (2006).

Separate regressions need symmetrization

At the population level, the two regression coefficients vanish together. In finite samples, however, we may have

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AND rule: Both regressions must select the edge.

$$\{j,k\} \in \widehat{\mathcal{E}}_{\text{AND}} \iff k \in \widehat{S}^{(j)} \text{ and } j \in \widehat{S}^{(k)}.$$

OR rule: One regression is sufficient.

$$\{j,k\} \in \widehat{\mathcal{E}}_{\text{OR}} \iff k \in \widehat{S}^{(j)} \text{ or } j \in \widehat{S}^{(k)}.$$

$\widehat{\mathcal{E}}_{\text{AND}} \subseteq \widehat{\mathcal{E}}_{\text{OR}}$: AND retains fewer edges for the same fitted regressions.

Local neighborhoods or a joint precision matrix?

Nodewise LASSO

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- Recovers a graph after AND/OR symmetrization.
- Does not directly supply a symmetric positive-definite precision estimate.

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Graphical LASSO

- Fits a single joint Gaussian model.
- Estimates a symmetric positive-definite precision matrix.
- Uses one objective combining likelihood and sparsity.

These are different estimators, motivated by the same Gaussian graph.

Gaussian observations and empirical covariance

Now allow an unknown mean:

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathcal{N}(\mu, \Sigma), \quad \mu \in \mathbb{R}^p, \quad \Sigma \succ 0.$$

The maximum-likelihood estimate of the mean is

$$\widehat{\mu} = \frac{1}{n} \sum_{i=1}^{n} X_i.$$

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The empirical covariance uses the likelihood normalization $1/n$:

$$\boxed{S = \frac{1}{n} \sum_{i=1}^{n} (X_i - \widehat{\mu})(X_i - \widehat{\mu})^\top = \frac{1}{n} D^\top D.}$$

$S \succeq 0$, but S need not be invertible. Centering gives $\operatorname{rank}(S) \leq \min(p, n-1)$.

From the Gaussian model to the likelihood

For $P = \Sigma^{-1} \succ 0$, the joint log-likelihood is

$$\ell(\mu, P) = \frac{n}{2} \log \det P - \frac{1}{2} \sum_{i=1}^{n} (X_i - \mu)^\top P (X_i - \mu) - \frac{np}{2} \log(2\pi).$$

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Set $\mu = \widehat{\mu}$. The identity $a^\top P a = \mathrm{tr}(a a^\top P)$ yields

$$\sum_{i=1}^{n} (X_i - \widehat{\mu})^\top P (X_i - \widehat{\mu}) = n\,\mathrm{tr}(SP).$$

Therefore

$$\boxed{-\frac{2}{n} \ell(\widehat{\mu}, P) = -\log\det P + \mathrm{tr}(SP) + p \log(2\pi).}$$

Course I: a probabilistic model defines an optimization criterion.

Why regularize the precision estimate?

Without regularization, minimize

$$f(P) = -\log\det P + \operatorname{tr}(SP), \qquad P \succ 0.$$

If $S \succ 0$, the first-order condition gives

$$-P^{-1} + S = 0 \quad \Longrightarrow \quad \widehat{P} = S^{-1}.$$

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- The inverse empirical covariance is generally dense and can be unstable.
- If S is singular, there is no finite positive-definite MLE: f decreases without bound along a null direction of S.

Sparsity regularization stabilizes estimation and selects graph edges.

Graphical LASSO: sparse penalized likelihood

$$\widehat{P} = \underset{P \succ 0}{\operatorname{argmin}} \left\{ -\log\det P + \operatorname{tr}(SP) + \lambda \|P\|_1 \right\}, \quad \lambda > 0.$$

Convention in this course: Entrywise ℓ_1 norm,

$$\|P\|_1 = \sum_{j,k=1}^{p} |P_{jk}| = \sum_j P_{jj} + 2\sum_{j<k} |P_{jk}| \quad (P \succ 0).$$

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- $-\log\det P + \operatorname{tr}(SP)$: Gaussian data fit.
- $\lambda\|P\|_1$: encourages off-diagonal zeros and penalizes the diagonal too.
- $P \succ 0$: ensures a valid Gaussian precision matrix.

The estimated graph uses the off-diagonal support of $\widehat{P}$.

The role of the regularization parameter

Small λ: More weight on fitting the empirical covariance. Estimated graphs are typically denser.

Large λ: More shrinkage and stronger sparsity promotion. Estimated graphs are typically sparser.

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Small λ: More weight on fitting the empirical covariance. Estimated graphs are typically denser.

Large λ: More shrinkage and stronger sparsity promotion. Estimated graphs are typically sparser.

- Too little regularization can retain spurious associations.
- Too much can remove meaningful conditional dependencies.
- The penalty depends on variable scales: scaling choices matter.

The trade-off concerns statistical fit and graph complexity. Individual edges need not disappear monotonically along the entire path.

Why is Graphical LASSO convex?

- The cone of symmetric positive-definite matrices is convex.
- $P \mapsto -\log\det P$ is strictly convex on this cone.
- $P \mapsto \operatorname{tr}(SP)$ is linear.
- $P \mapsto \|P\|_1$ is convex, but nonsmooth.

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- $P \mapsto \|P\|_1$ is convex, but nonsmooth.

For a symmetric direction $H \neq 0$, the log-determinant curvature is

$$\mathrm{d}^2(-\log\det)(P)[H,H] = \operatorname{tr}(P^{-1}HP^{-1}H) = \|P^{-1/2}HP^{-1/2}\|_F^2 > 0.$$

With $S \succeq 0$ and $\lambda > 0$, the full-entry penalty gives an attained, unique minimizer, even if S is singular.

ADMM (1/5): separating likelihood and sparsity

Variable splitting: Introduce a symmetric matrix Z:

$$\min_{P \succ 0,\, Z = Z^\top} f(P) + \lambda \|Z\|_1 \quad \text{subject to } P = Z.$$

Here $f(P) = -\log\det P + \operatorname{tr}(SP)$.

P handles Gaussian likelihood and positive definiteness. Z handles entrywise sparsity.

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Scaled augmented Lagrangian: With $\rho > 0$ and $U = Y/\rho$,

$$\mathcal{L}_\rho(P, Z, U) = f(P) + \lambda \|Z\|_1 + \frac{\rho}{2}\|P - Z + U\|_F^2 - \frac{\rho}{2}\|U\|_F^2.$$

Here Y is the ordinary Lagrange multiplier and $\|A\|_F^2 = \sum_{j,k} A_{jk}^2$.

Alternate minimization in P and Z, then update U to enforce agreement.

ADMM (2/5): the positive-definite precision update

$$P^{t+1} = \operatorname*{argmin}_{P \succ 0} \left\{ -\log\det P + \operatorname{tr}(SP) + \frac{\rho}{2}\|P - Z^t + U^t\|_F^2 \right\}.$$

The first-order condition is

$$\rho P - P^{-1} = \rho(Z^t - U^t) - S =: A^t.$$

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Diagonalize the symmetric matrix $A^t = Q\operatorname{diag}(d_1, \ldots, d_p)Q^\top$. In this basis, each eigenvalue q_i of P^{t+1} satisfies

$$\rho q_i - q_i^{-1} = d_i \quad \Longrightarrow \quad q_i = \frac{d_i + \sqrt{d_i^2 + 4\rho}}{2\rho} > 0.$$

$$\boxed{P^{t+1} = Q\operatorname{diag}\left(\frac{d_i + \sqrt{d_i^2 + 4\rho}}{2\rho}\right)_{i=1}^{p} Q^\top.}$$

A spectral update solves the likelihood subproblem and guarantees $P^{t+1} \succ 0$.

ADMM (3/5): the sparse-variable update

For $B^t = P^{t+1} + U^t$, the second minimization reads

$$Z^{t+1} = \underset{Z=Z^\top}{\operatorname{argmin}} \left\{ \lambda \|Z\|_1 + \frac{\rho}{2} \|Z - B^t\|_F^2 \right\}.$$

Each entry solves an ℓ_1 proximal problem. Since B^t is symmetric, entrywise thresholding also preserves symmetry:

$$\boxed{Z_{jk}^{t+1} = \mathrm{soft}_{\lambda/\rho}(B_{jk}^t)}, \qquad \mathrm{soft}_\tau(b) = \mathrm{sign}(b) \max(|b| - \tau, 0).$$

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All entries are thresholded, including the diagonal. An off-diagonal-only penalty would require a different diagonal update.

Course II: the same soft-thresholding operator promotes sparsity. Z^{t+1} is symmetric, but need not be positive definite.

ADMM (4/5): agreement and stopping

After the two minimizations, update the scaled multiplier:

$$\boxed{U^{t+1} = U^t + P^{t+1} - Z^{t+1}.}$$

U accumulates violations of the constraint $P = Z$.

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Two residuals to monitor:

$$r^{t+1} = P^{t+1} - Z^{t+1}, \qquad s^{t+1} = \rho(Z^{t+1} - Z^t).$$

Stop when both Frobenius norms are below chosen tolerances.

- $\|r^{t+1}\|_F$: agreement between the two matrix estimates.
- $\|s^{t+1}\|_F$: dual residual, measuring the change in Z.

λ defines the estimator. ρ controls the algorithm's behavior, not the converged minimizer.

ADMM (5/5): complete Graphical LASSO algorithm

Inputs: $S \succeq 0$, $\lambda > 0$, fixed $\rho > 0$. Initialize $Z^0 = I_p$, $U^0 = 0$. For $t = 0, 1, \ldots$:

1. **Likelihood and positive definiteness:**

$$A^t = \rho(Z^t - U^t) - S = Q\mathrm{diag}(d_i)Q^\top, \qquad q_i = \frac{d_i + \sqrt{d_i^2 + 4\rho}}{2\rho},$$

$$P^{t+1} = Q\mathrm{diag}(q_i)Q^\top.$$

2. **Sparsity:** $Z^{t+1} = \mathrm{soft}_{\lambda/\rho}(P^{t+1} + U^t)$ entrywise.
3. **Agreement:** $U^{t+1} = U^t + P^{t+1} - Z^{t+1}$.

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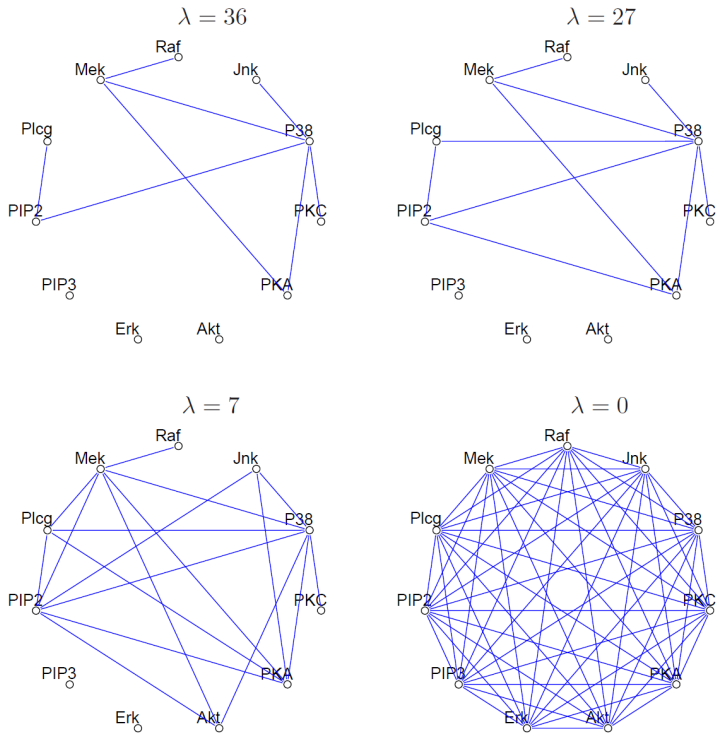
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2. **Sparsity:** $Z^{t+1} = \mathrm{soft}_{\lambda/\rho}(P^{t+1} + U^t)$ entrywise.
3. **Agreement:** $U^{t+1} = U^t + P^{t+1} - Z^{t+1}$.

Stop using both residuals. With exact updates, P^t and Z^t converge to the same GLASSO solution.

At finite tolerance, use P^t for a positive-definite matrix; Z^t exposes sparsity. Check their agreement before extracting the graph. Boyd et al. (2011), Section 6.5.

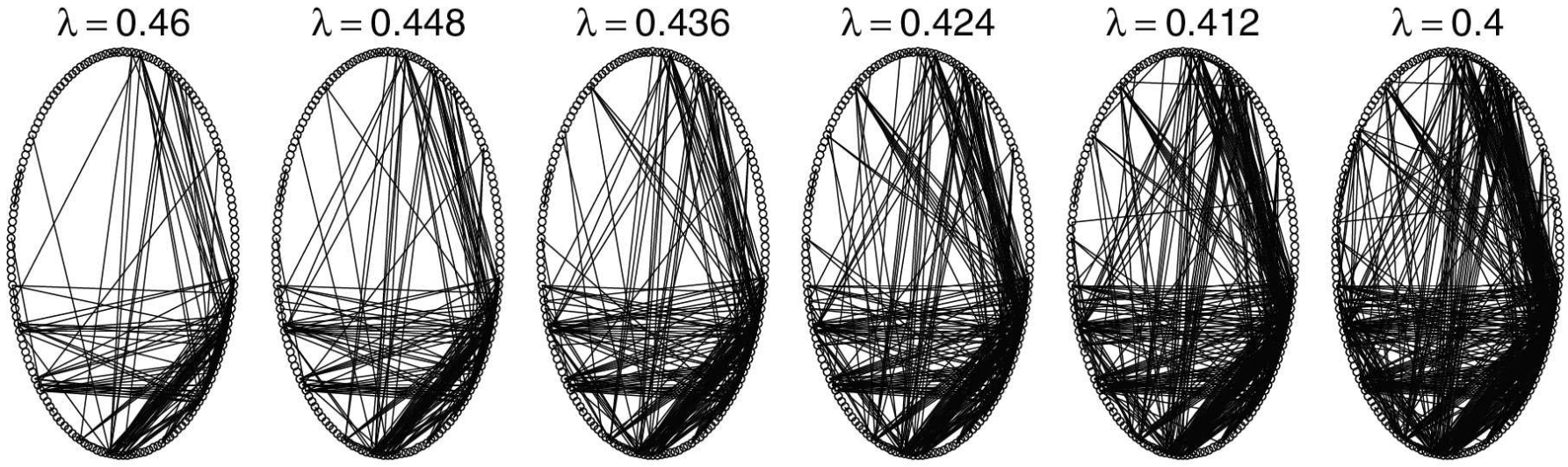
Flow-cytometry data



Four GLASSO fits for $p = 11$ proteins measured on $n = 7466$ cells. Sachs et al. (2005).

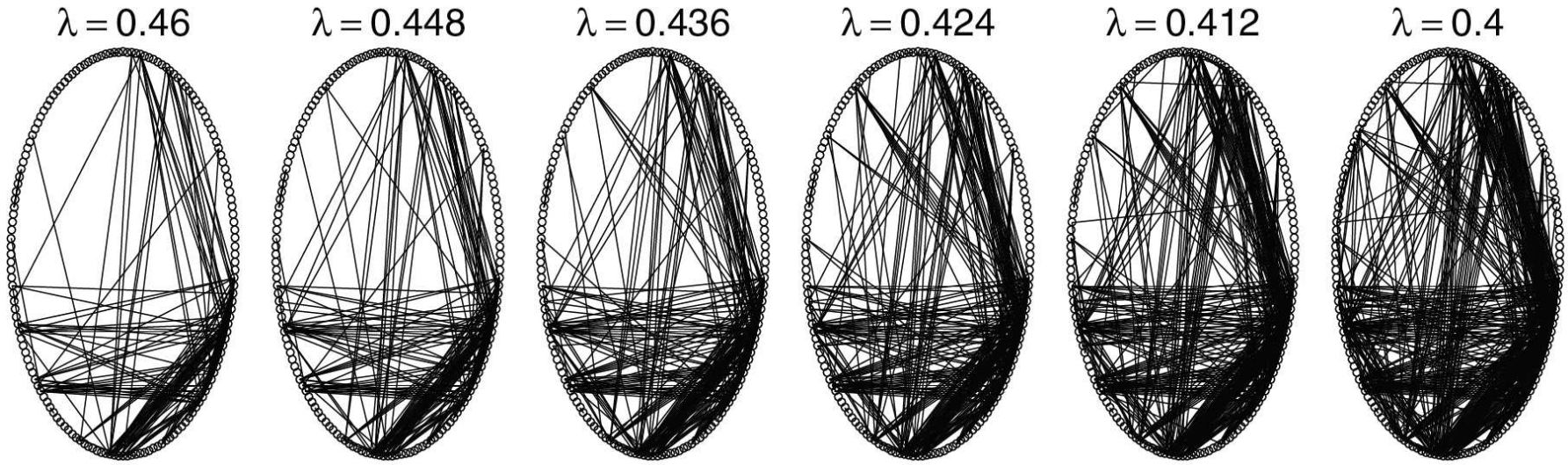
The displayed graphs become denser as λ decreases. Edges represent conditional associations under the Gaussian model.

Riboflavin genomic data



Six GLASSO fits for the riboflavin example with *Bacillus subtilis*: $p = 160$ randomly selected genes and $n = 115$ samples. Meinshausen and Bühlmann (2010).

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Six GLASSO fits for the riboflavin example with *Bacillus subtilis*: $p = 160$ randomly selected genes and $n = 115$ samples. Meinshausen and Bühlmann (2010).

- Here $p > n$: the empirical covariance cannot be inverted.
- Smaller displayed λ values retain more connections.

Regularization makes joint estimation possible and gives a graph whose complexity can be controlled.

Connection to Lab 6

The existing lab compares two graph estimators:

- Graphical LASSO solved with ADMM.
- Nodewise LASSO with AND and OR rules.

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Questions to investigate:

- How does the sample size affect edge recovery?
- How do AND and OR change accuracy and F1 score?
- How accurately does GLASSO estimate the precision matrix?

Separate graph recovery from matrix estimation: a good edge pattern and a small matrix error are different objectives.

Inference on graphical models: take-home messages

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- Graphical LASSO jointly estimates a sparse positive-definite precision matrix.

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- Regression coefficients give local neighborhoods. Nodewise LASSO uses variable selection and symmetrization.
- Graphical LASSO jointly estimates a sparse positive-definite precision matrix.
- ADMM separates the likelihood update from soft-thresholding, and enforces agreement through the dual variable.

Probability, regression, sparsity and convex optimization combine to recover an interpretable statistical structure.

Selected references

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